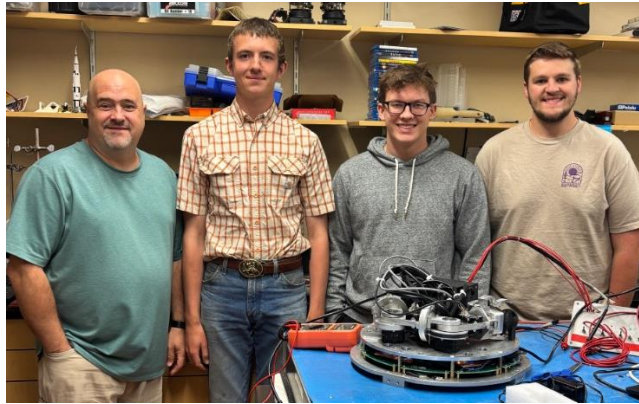


RockSat-X 2025 Final Report

NNU RockSat

To further the study and exploration of microgravity robotics by using a semi-autonomous robotic system to launch, track, and retrieve a ball ejected into space.



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1.0 Mission Statement

Intelligent robotic systems have the potential to assist astronauts and solve challenging problems such as space debris removal, collision avoidance, and in-orbit spacecraft servicing and assembly. Current robotic systems used in space require manual control, which will prove difficult for distant uncrewed missions where communication times make remote control impractical. Remote control will also hinder the creation of large robotic workforces, as the necessity for human involvement would be too great. This technology could be revolutionary in the aerospace industry as an intelligent, dexterous system could learn to perform tasks in space and would decrease the need for human intervention. This work aims to develop adaptive robotic arms that train in simulation and then complete a complex task in the real-world space environment. For this project, the complex task involves one robotic arm that catches a ball in space.

2.0 Mission Requirements and Description

The NNU RockSat team has had a long history of attempting autonomous microgravity robotics, and the 2024-2025 design was an iteration of the previous years' work. The project consisted of a robotic arm that was capable of performing a given task in microgravity. A series of objects would be ejected from an autonomous deployment system that launched balls into space, and then a depth camera mounted within the fingers of the robotic arm would track the ball. When the ball is successfully tracked, the robotic arm would move into position and grab said ball, successfully completing the mission.

The team decided to once again use a half-cam instead of a whole-cam system, as per the previous year's experiment. The design constraints also remained the same, with a dry box, a singular robotic arm, and a ball launcher. All of the subsystems fit comfortably within the usable volume, and the deployment method was specifically focused on to strive for a safe and highly predictable trajectory.

Other requirements included asking for the removal of part of a longeron and an additional request for Ah that had to be approved by Wallops. The vision system, while iterated from the previous years' projects, needed an overhaul, and the model had to be coded to utilize its given training to recognize red balls in any background. A slight concern was found here, which was discussed in the pointing request. The vision camera as well as the GoPro (used to document the project's functionality in microgravity) needed a view of the sun in order to fully capture a vision of the project. The object ejector had the necessary constraint to eject the object in less than 1 in/s. Other constraints worthy of mention were that each joint would have 150% of the torque required to operate in gravity when operating in space.

The system itself had a few specific desirables that were sought. The robotic arm needed to be capable of moving the end effector to a precise position, and while it was necessary to be extended to run the mission, before re-entry, the extended arm would be retracted and secured. The software written would also need to be capable of locating the desired object in RGBD footage.

3.0 Experiment Design

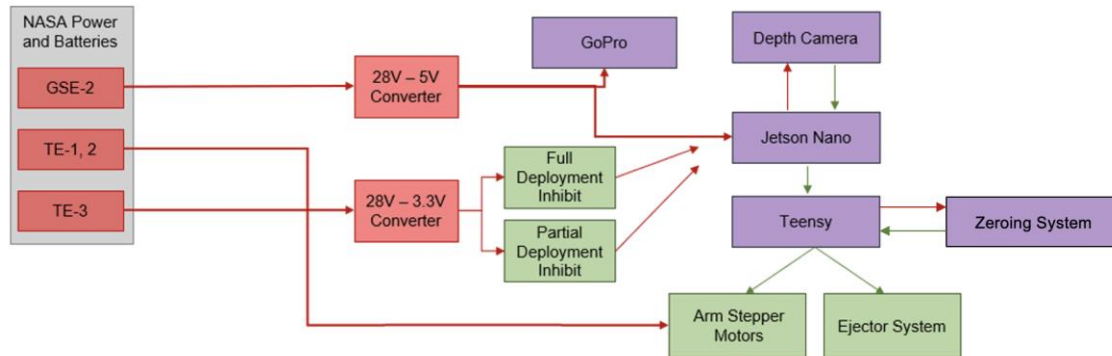


Figure [1] - Functional Block Diagram

DAEDALUS IV Design Overview

Our design featured three main components: the electrical system providing power and data lines to the various subsystems, the mechanical structures holding everything together, and the arm/ejector system. We were able to reuse most of the mechanical structures from last year's project, although we did have to mirror them in order to meet design requirements. We redesigned the power system, combining several daughterboards into a single PCB and adding an integrated inhibit system. We also had to completely redesign the ejector system to eliminate the use of magnets. The arm assembly remained effectively identical to last year's design.

We were well within our projected power budget, and about 1 lb under on our weight budget, which was rectified by a special allowance from the Wallops technicians.

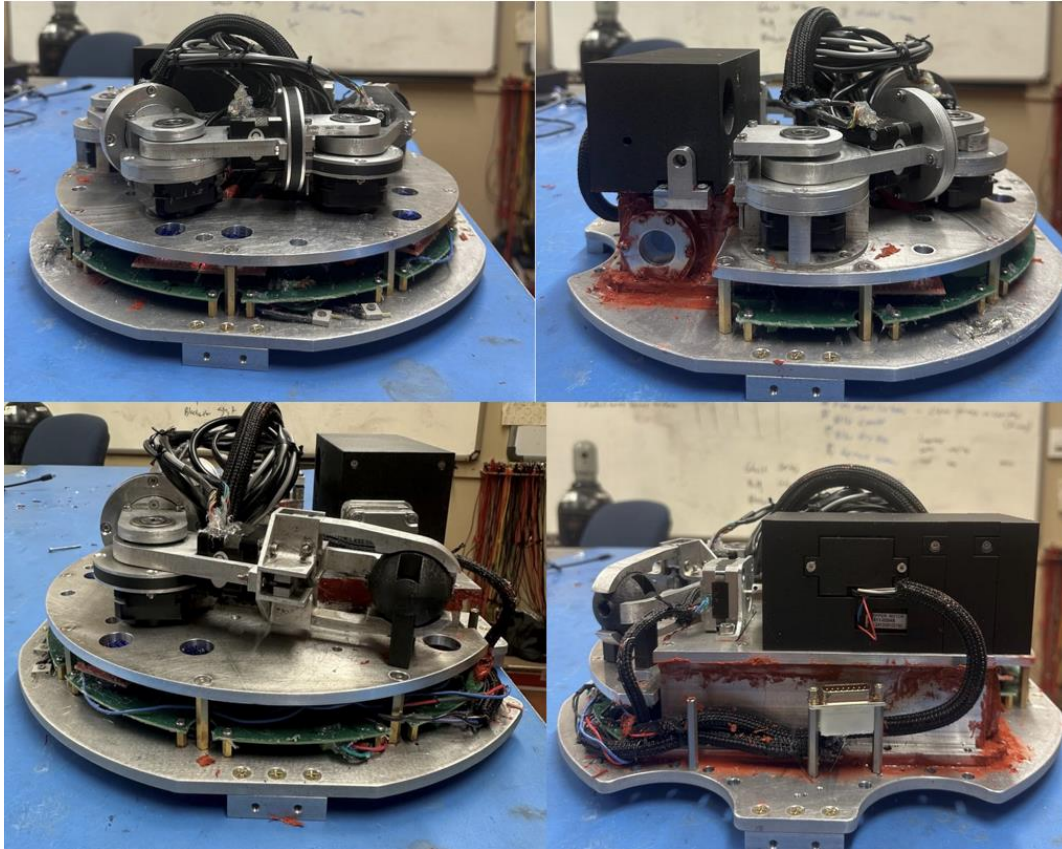


Figure [2] - Orthogonal Views Pre-Launch

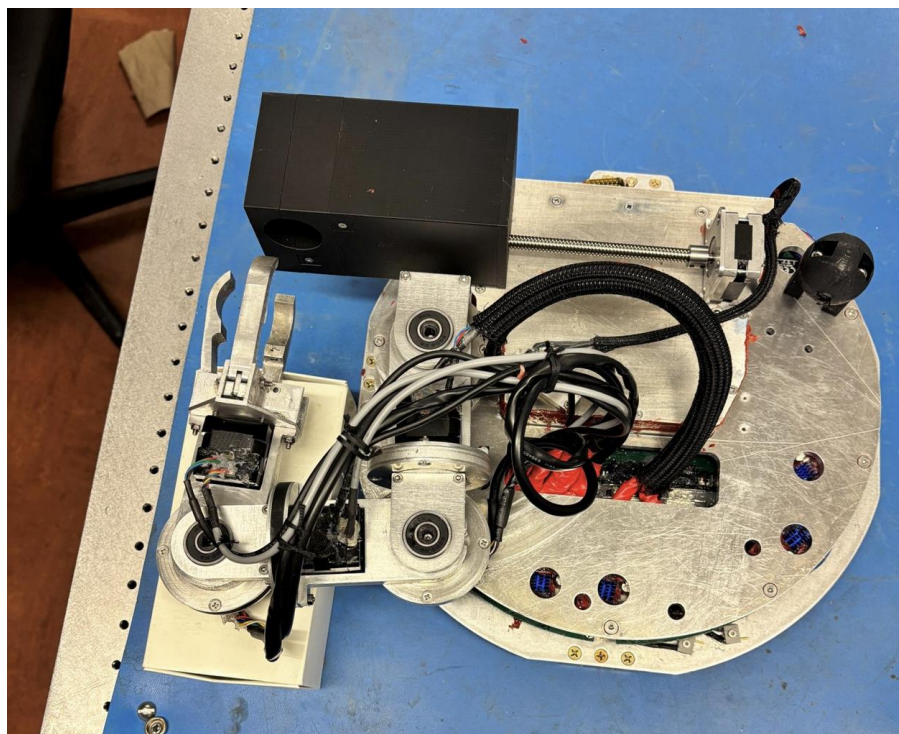


Figure [3] - Arm and Launcher Extended Pre-Launch

Electrical System Redesign

As part of the electrical system simplification process, many external boards that were implemented last-minute from the Daedalus II mission were combined into a single PCB. These boards include the inhibit pin system and the driver for the ejector motor. Our board was designed using AutoDesk Fusion 360.

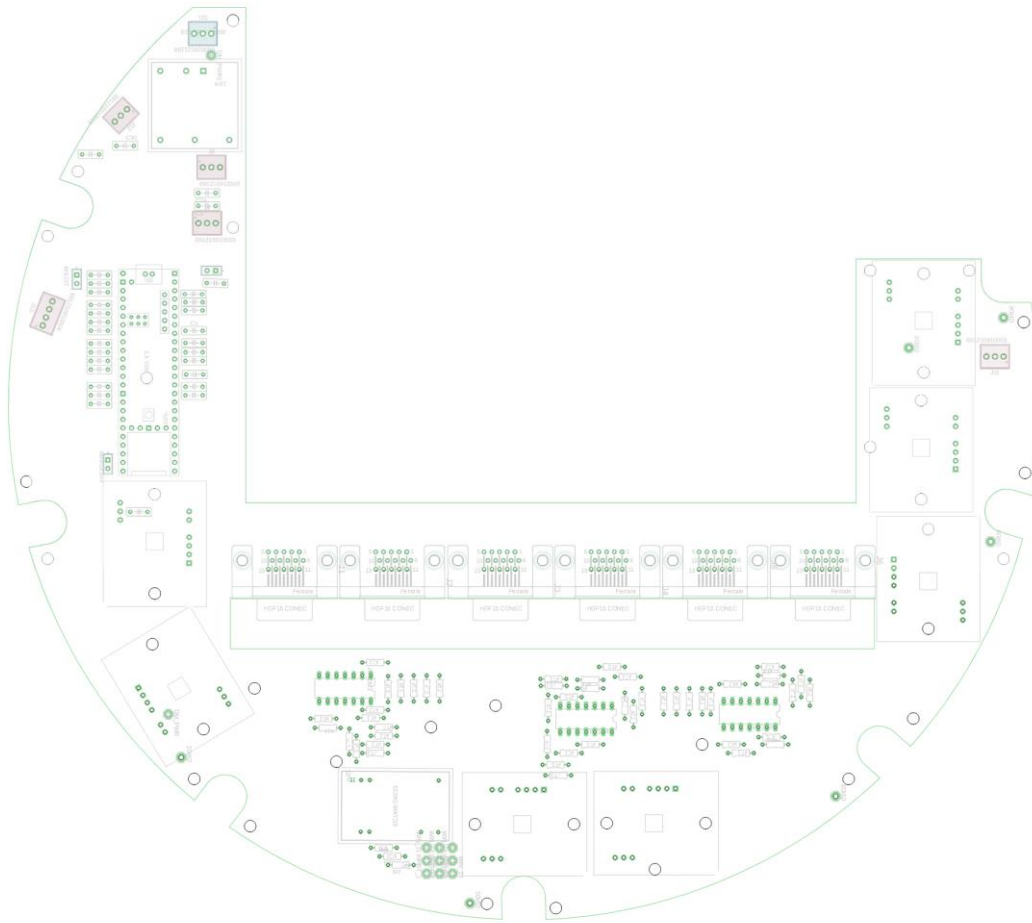


Figure [4] - PCB Schematic

Features of the board include test pads, where the voltage of power traces can be measured, a 9-pin inhibit circuit, and capacitor mounts, where a capacitor could be soldered to reduce noise and interference in the power lines. After rigorous testing, it was determined that no capacitors were needed to reduce noise. Header pins were used to connect both the BigEz stepper motor controllers and Teensy to the board. This feature allowed for quick replacement of microcontrollers in the event of a malfunction. Similarly, 9-pin D-Sub connectors were used to connect individual stepper motors and their respective encoders to the microcontroller. This allows for easy replacement of motors should an encoder burn out.

Power was distributed to the board by the following lines: GSE, TE-RA, and TE-RB. The 28V GSE line was stepped down to 5V by a voltage converter, which powered the Jetson, Teensy, and GoPro. The TE-RA and TE-RB lines went straight to the 8 Big Easy stepper drivers.



Figure [5] - Power Into Board

The Teensy is mounted at the bottom of the board. When a movement command is sent from the Jetson, the Teensy determines the desired speed using a PI controller. A signal is sent to the BigEz microcontrollers and then to stepper motors through the D-Sub connectors. As the motor moves using an op-amp circuit, the encoder values from the four redundant lines are compared, and then the value is sent to the Teensy.

A second circuit was developed to control the inhibit system. The circuit diagram in Figure 6 outlines the inhibit circuitry. TE-3 comes into a 3.3V converter, which powers Winchester pins that were used to create breaks in the circuit. When a low was read on the GPIO board of the Jetson, the associated inhibit procedure was performed.

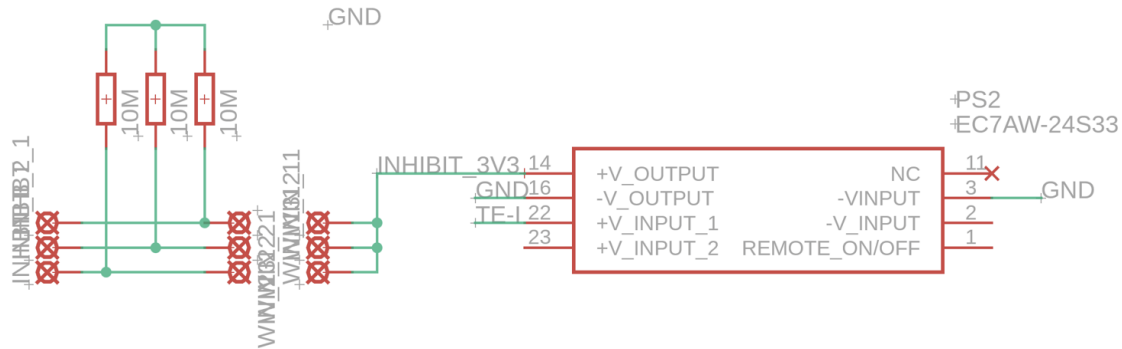


Figure [6] - Inhibit System

Payload Command System Software

The payload requires an overhead system to control each of the subsystems. Named the payload command system (PCS), this system is executed on the main computer, a Nvidia Jetson. The PCS is a ROS service in which each payload subsystem queries to indicate its current state and to receive commands such as activation, deactivation, and shutdown.

Outlined in Figure 7 the blue lines indicate a success or presence of multiple choices while orange lines indicate a failure. When GSE is activated, the Jetson Nano receives power and begins to boot up. During the bootup process, the PCS begins as a system service. At the beginning of the PCS, a check is run to determine the status of the TE lines. The system waits until this signal has been received, switching between the TE detection state and the wait for the TE delay state until the signal is received.

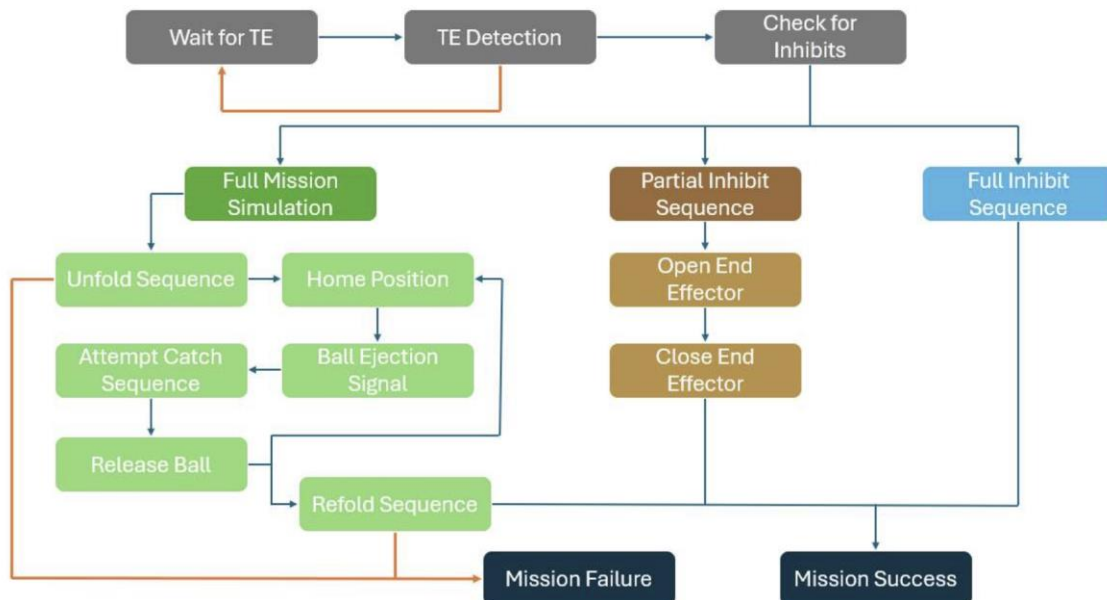


Figure [7] Master Payload State Machine

At T+110, the first TE signal is sent, informing the experiment that the rocket has spun down. Once the first TE event is registered, the system checks for inhibit signals. The PCS system detects the inhibit board procedure using the Jetson's GPIO board. Depending on the combination of high and low signals, one of the three procedures is enacted. If the payload is fully inhibited, the system will exit, and all processes will cease. At this time, the stepper motors maintain their position and wait for power shutoff. If the partial procedure is called, the payload latches and unlatches to its fixed position on the payload. The combinations of these two tests allow the team to test the functionality of all arm joints without running a full mission sequence.

If the full mission sequence is called, the arm unlatches from its mount and begins unfolding; first, it zeros its position, then it attempts to unfold to the home position. Once the home position is reached, the arm will begin the catch sequence. A signal is sent to the ball launcher to eject a ball. After a delay, the arm will attempt to catch the ball.

This process is repeated for each deployable, and then the arm refold sequence is called. This sequence steps through the unfolding sequence's positional code backward and ends by latching to the fixed position.

Camera System Design

The camera system for this experiment consisted of a single GoPro Hero 11 mounted next to a viewport inside the dry box. Its purpose was to record footage of the arm attempting to catch the ejected ball. Using the GoPro Labs firmware, QR codes were generated that told the GoPro to start recording upon receiving power, and to save the video every 15 seconds to avoid video corruption issues. The GoPro was powered through a line on the PCB, eliminating the need for any batteries adding unwanted stored energy.

Computer Vision System Design

Originally, the project was to contain an Intel Realsense D405 running the YOLOv8 image recognition model to allow the arm to dynamically track and capture the ball. However, due to deadline constraints, this system was de-scoped.

Ball Ejector Design

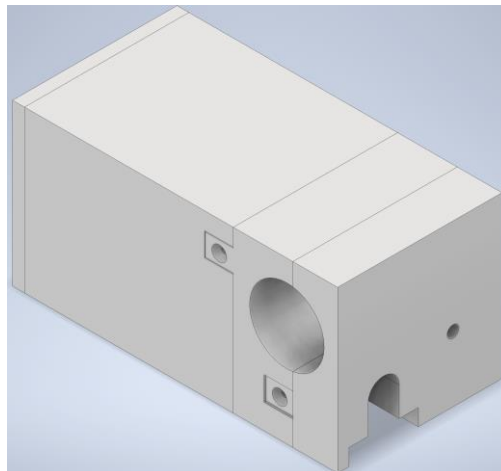
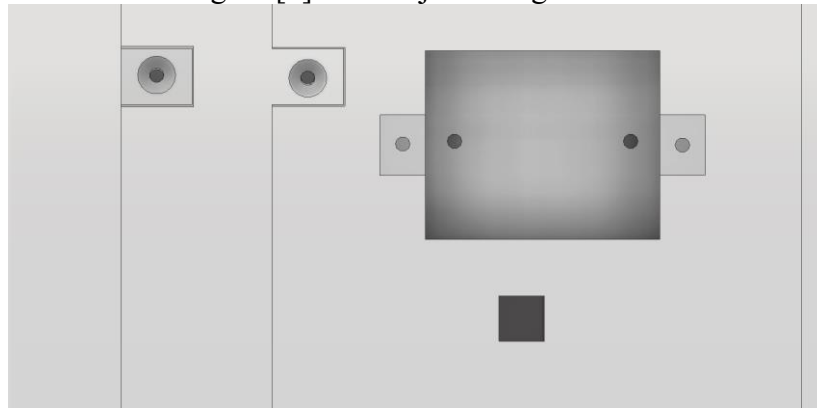


Figure [8] - Ball Ejector Isometric View

Due to the need to eliminate magnets from our mission, a purely mechanical ball ejector was designed. It consisted of a main component extended out of the body of the experiment by a threaded shaft. Once it was in place, a spring and system of cams ejected the balls one by one.

Figure [9] - Ball Ejector Right View



On the right side of the ejector, there is an embedded slot for a Big Easy driver to drive the cam motor inside. Each component of the ejector shell is connected by tabs with a screw driven through it.

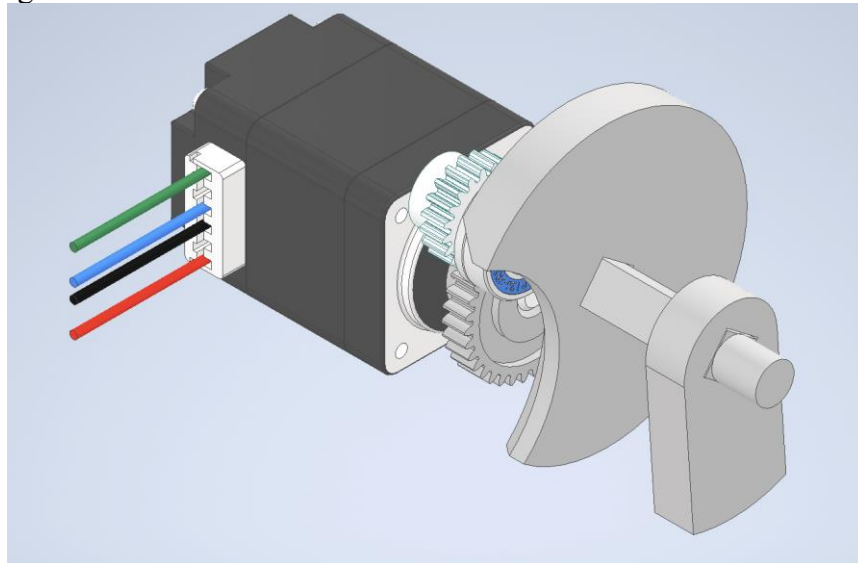


Figure [10] - Isometric View Cam System

The cam system held back all the balls when the round cam was in place, but when it rotated it let one ball by, then the secondary cam gave it a whack and sent it out. The rotation speed of the motor could be controlled to modify the ejection rate.

Robotic Arm Design

We used an arm system very similar to what was used on previous missions, except that we used 3-D printed gears to eliminate the friction caused by the waterjet cut gears of previous missions.

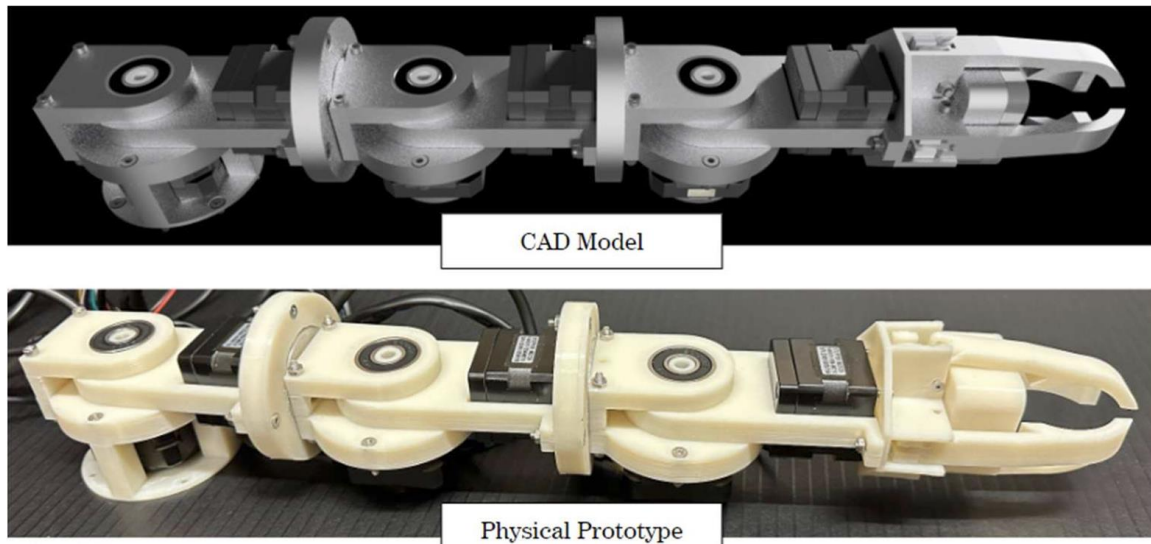


Figure [11] - Arm Design and Actuation

The arm was made out of aluminum for the mission, but our prototype was made of PLA. The arm consisted of 3 stepper motors controlling movement in the xy plane, and two controlling movement in the z plane. An additional motor at the end of the arm controlled the gripper using an ACME threaded screw to open and close.

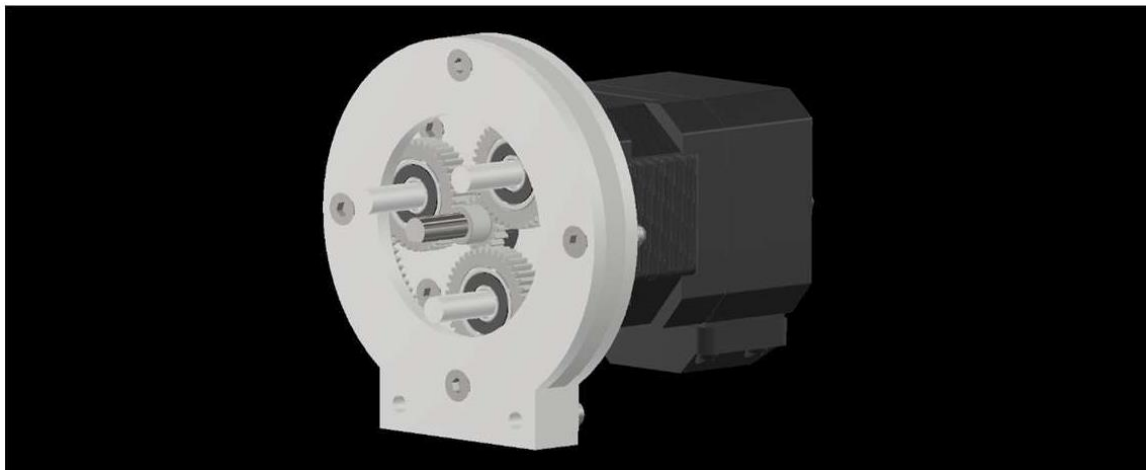


Figure [12] - Planetary Gearbox CAD Model

Due to the high amount of expected torque, we used the same planetary gearboxes for each joint. Because we use waterjet cut parts, the tolerance on our gears wasn't close enough for smooth rotation. So, we switched to using glass-fiber and ABS printed gears inside the gearboxes to reduce friction. The arm secured itself by gripping to a mounting point on the midplate.

4.0 Student Involvement

This project was a completely student lead program. The students were in charge of designing, prototyping, and testing the project. This involved all of the team management, time management, and decision making involved. A major part of this year's student involvement was centered around the pass off of the project from leaving seniors to incoming members. Following is a list of the most involved peoples.

Ethan Buchanan (Mechanical Engineering, Exiting Team Lead): Ethan was heavily involved in the initiation of all parts of the project, and the passing off of knowledge to new members.

Josh Nelson (Computer Science, Exiting Computer Vision Code Lead): Josh was tasked to develop a machine learning image recognition program to recognize and locate the ball using our specific depth camera.

Elias Friberg (Computer Engineer, Incoming Team Lead): Elias was tasked with taking over team communication with NASA, and general team management. Additionally, he was Computer and Micro-Electronics Lead.

Ben Yantes (Electrical Engineering, Mechanical Lead): Ben handled all the CAD models and preparation of files for our machinist to manufacture, as well as the assembly and troubleshooting of the arm.

Jonathan Stillman (Engineering Physics, PCB Lead): Jonathan soldered up the PCB and handled all the electrical connections between the various subsystems, as well as the integration of the Vision System with the project.

McKenzie Fitzgerald (Engineering Physics): McKenzie handled presentation compilation and software logging capabilities.

5.0 Testing Results

Robotic Arm System Testing

Each joint in the arm was powered off of a Big Easy stepper driver. Each driver had a potentiometer on it to regulate current. Each joint's potentiometer was adjusted to give it enough torque to move the arm effectively. During initial testing, it was determined that the driver boards were drawing too much current and overheating, which caused chattering in the gearboxes. We had to glue on heatsinks in order to fix this problem. We simulated microgravity by holding each joint up with a string when trying to move each joint, and the arm was reliably able to move around, though not particularly smoothly. We also discovered upon assembly of the aluminum arm that the inner and outer races on our bearings were pressing against the same surface, somewhat defeating their purpose. We had to shrink the diameter of the inner rod going through the bearing so it could spin freely. We also ran into problems with the planetary gears rubbing too much on the top and bottom of the casing. We made the outer race thicker and added a spacer under the gears to eliminate most of these friction problems.

Ball Ejector System Testing

When testing the ball ejector on a breadboard, it worked really well. However, upon integration with the main project, we realized that the Big Easy Stepper driver (the type of driver used for all motors) was overdriving the motor. For a short test run on the breadboard, this was unnoticeable, but the full mission tests with the motor encased in PLA revealed the problem. Unfortunately, because other subsystems had a higher integration priority, this problem wasn't realized until it was too late to resolve.

Because the motor heated up so much and the cam system had a lot of heavy metal gears, the PLA shaft powering the cam system that was press fit onto one of the gears kept melting and would stop turning, rendering our ejector unusable. Again, because we discovered this problem too late, we weren't able to come up with a satisfactory solution. These problems lead us to decide to descope the actual ejection of the PLA ball. The launcher still extended, it just didn't launch anything.

Camera System Testing

During assembly testing, we realized that the drybox, which had been mirrored from last year's project, no longer was correctly oriented for the GoPro. During the mirroring process, everything was flipped, but they don't make flipped GoPro's. So, our viewport lens and our GoPro lens did not line up correctly. To fix this, our machinist cut a slot in the drybox with another viewport on the bottom, and we flipped the GoPro over to see through the port.

We used the GoPro Labs software to program it using scannable QR codes.

Vision System Testing

During final assembly, when we covered all the components in RTV and sealed the drybox, something happened to the USB cable going from the Jetson to the depth camera. They were communicating fine before integration, but afterwards they were not. We

presume this happened because we had to cut the sheathing off the cable to get it to fit in the drybox, and we must have nicked a data line. Because everything was already covered in epoxy, we weren't able to fix it, and had to descope the camera.

Electrical System Testing

Considering the electrical aspects of the project, there was a lot that needed to be debugged due to lack of passed down knowledge of the project and lack of education in microelectronics. The biggest problem we had was a situation where a problem was believed to exist in reading in motor potentiometers, when, in reality, the problem was really caused by the testing environment being incorrectly set up. Another involved a problem where our driver boards fed 5V into the Teensys' GPIO ports, effectively frying the microcontroller. The solution to this problem was a solder bridge point on the motor drivers that set them to interface with the Teensy at 3.3V instead of 5V. Learning about and solving these issues were critical to our team's understanding of the system and how it worked.

One of the outcomes of the testing of the electrical systems was a lasting voltage on the TE-R pins of the project. This was a concern at first until it was determined that this was simply residual charge from the capacitors in the driver boards. Another concerning aspect of final electrical testing was the discovery of some voltage on the baseplate. After discovery, a full mission simulation was performed to log the stray voltage over the course of the entire mission. After doing this, the voltage was deemed minimal and constant enough that it was safe to ground the baseplate, which did reduce voltage to a negligible amount.

6.0 Mission Results

Due to causes outside of our control, our project was not recovered. We don't know the full reason yet, but we do know that our project, which was on the bottom of the payload, snapped off (supposedly on impact with the water after reentry). As such, we didn't recover any data. The Wallops technicians said that this was the first time in the history of the project that anything like it had happened.

During testing at Wallops, however, we realized that our GoPro was programmed to start recording on power on of the GSE line, and that every test they ran was filling up the SD card. By our calculations, we were right on the edge of not having any room left for mission video by the time all the testing was complete. Also, because our inhibit pins were all at the software level, all of our subsystems were powered on, though incapable of movement, which drained the rocket's batteries a lot faster than desired.

7.0 Conclusions

We didn't get any data from our mission, but we did learn that a system to toggle power on and off for each individual system would both prevent excessive drain on NASA's batteries as well as prevent the SD card on our GoPro from filling up during testing.

After writing this review, we are releasing how easily we could be disappointed with how this year of RockSat turned out. There have been countless issues and roadblocks, many things were de-scoped before integration, and of course, after all was said and done our results equated to some sheared bolts. Yet, that is not our outlook on this experience at all. For every single thing that got de-scoped, there were countless times where we were able to build out the functionality of something that we previously had no idea about how it worked. For every stressful late night working on our project there is a pile of gratitude for the chance to be able to put our work into space. And even though we were unable to retrieve our data, we still got to experience the RockSat-X Virginia trips, we still get to say we've worked with NASA, and the entire experience has made us even more excited to do it all again this upcoming year. Despite all these setbacks we can guarantee that we grew especially as engineers throughout this experience, and it has forever positively affected our life's trajectory.

8.0 Potential Follow-on Work

Due to the lack of experience of the incoming team (who did the majority of the project), and the hopes of various improvements we want to make, we are planning to continue iterating the mission for this next year. An example of these improvements is our plan to implement a more functional inhibit system (that allows us to control each subsystem's access to power, dependent on subsystems). This will solve our concerns of the GoPro's storage filling up, as well as prevent excessive draw on NASA's batteries during testing. This could also enable better logging of the subsystems' various voltages and current draws.

Further follow up work would include the design of a better ball launcher and arm. Because our ball launcher was made out of PLA, many components melted or warped due to the heat from the stepper motors inside. A more optimized design would allow critical parts to be machined out of metal, as well as improved ventilation for the motors, preventing them from getting too hot.

Our current arm design is unreliable at best. This is because the tolerance in the joints had to be very loose in order for the gears not to bind up. Loose tolerance corresponds to unpredictable arm movement, which causes all sorts of problems. A redesign of the arm to make each joint more precise would greatly improve its viability.

9.0 Benefits to the Scientific Community

The main benefit we hope this mission provides to the scientific community is in the field of space junk removal. Generally, this task is exceptionally hard, because there are so many unknowns involved in space junk removal missions. It's virtually impossible to program in every conceivable scenario before launch. So, a mission with a small form factor that can dynamically learn and respond to different situations is highly desirable because it can react to the unknown autonomously, without human control.

Our mission laid the groundwork for further testing of robotic arm maneuvering and machine-learning assisted vision in microgravity environments. Due to the loss of our mission, we didn't recover any usable data. However, through the design and assembly process, we learned a lot that will be very useful for upcoming missions. For example, we learned that we need a power interrupt system and a better arm and launcher system. With this knowledge, we can build a better project next year that will be much more successful and achieve far more in the future.

As we advance and develop this project further, we could conceivably end up with a fleet of space-cleaning robots operating autonomously. Ultimately, hundreds of our robots will be orbiting the earth, sweeping up mountains of space junk and throwing it into the atmosphere for incineration.

10.0 Lessons Learned

The biggest lesson we learned was that early preparation is really important. We were working on most presentations until the very last minute, and most of our designing was really rushed. This was mostly caused by a weird software bug that wasn't resolved until March, which really delayed the testing of the other components until very late in the semester. As such, we ran into the majority of our mechanical issues far too late to fix. Had we gotten an earlier start in the year, and found the problem in November, it could have been resolved before Christmas break and we would have had an entire semester for design optimization rather than a single month. Additionally, because we were so late, many critical subsystems had to be descope, like the depth camera and ball launcher. Had we assembled everything even a week or two earlier, we would have had time pre-VVC to fix the depth camera and ball launcher.

Another big thing we learned was that project handoff is really important. Many of the subsystems on our project were previously designed by people who have since graduated. Because they graduated and moved away, we couldn't ask them questions about how things worked or why they were designed the way they were. A lot of time was spent figuring everything out the hard way, when it could have been much smoother with clear documentation and handoff. This also contributed to the lateness of our project completion, because several systems had to be redone due to minor details that were missed because of inexperience.

On the more technical side of things, we learned that shafts printed out of PLA don't hold up really well when gears press fit onto them get really hot. In the future we would recommend the usage of steel shafts and bearings when possible. We also learned that a power interrupt system for the different subsystems would save everyone a lot of headaches. Being able to shut power to the GoPro off to prevent it from filling its SD card up during testing, as well as being able to prevent drain on the mission batteries would be really helpful.

11.0 Appendices

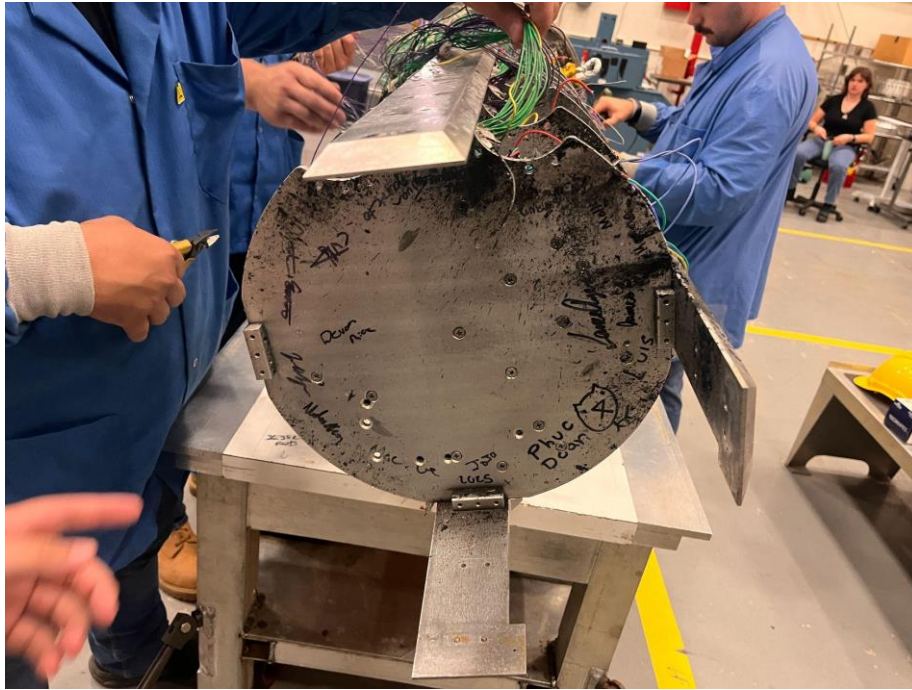


Figure [13] - NNU's Project Missing from Recovered Payload